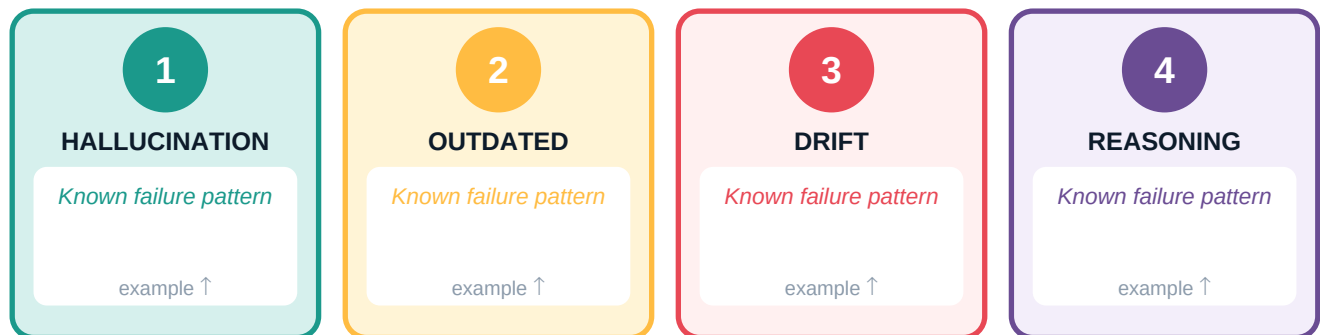


CHAPTER 21

What AI Gets Wrong (Consistently)



THE FAILURE TAXONOMY AI failures are not random. They fall into predictable categories that appear consistently across different models, different platforms, and different types of task. Knowing these categories means you can anticipate where to look, rather than discovering errors after they've caused a problem.

This chapter is a practical catalogue of known AI failure modes, not to make you sceptical of AI, but to make your scepticism accurate: applied where the risks are highest, relaxed where the risks are lower.



THE FAILURE TAXONOMY

01 Hallucination 02 Outdated information

03 Confident over- 04 Arithmetic errors generalisation

05 Context drift 06 Sycophancy

07 Instruction drift 08 Plausible but wrong reasoning

FAILURE CATEGORY 1, HALLUCINATION What it is: AI generates specific facts, names, dates, statistics, citations, or events that did not happen or do not exist, but presents them with the same confidence as accurate information.

Why it happens: AI generates plausible text, not verified text. A hallucinated statistic is generated because it is the kind of statistic that would plausibly appear in this type of document. A hallucinated citation is generated because it is the kind of citation that would plausibly appear in academic writing on this topic.

Where it appears most:

Citations and references (highly specific hallucinations) Statistics and percentages (plausible but invented figures) Biographical facts about real but less prominent people Historical detail about events
AI has limited training data on Company-specific information AI wasn't trained on

How to catch it: Verify specific claims against primary sources. Treat all citations as unverified until confirmed. Be especially sceptical of very specific numbers.

The severity: High for professional use. A hallucinated citation in a published article, a made-up statistic in a board presentation, or an incorrect legal date in a contract review, these cause real professional and sometimes legal consequences.

FAILURE CATEGORY 2, OUTDATED INFORMATION What it is: AI presents information that was accurate as of its training cutoff but is no longer current.

Why it happens: AI's knowledge is frozen at the point of training. The world continues to change. Models released in 2025 may have training data that lags by twelve to eighteen months before release, and then be used for years after.

Where it appears most:

Technology product details (features, pricing, availability) Regulatory and legal information
Organisational information (who holds which role, company structure) Market data and economic statistics
Current events and recent developments

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How to catch it: For any fast-moving topic, verify current information from official sources. Even with web search enabled, verify important current-state claims.

The severity: Medium to high, depending on context. Outdated legal information is high-severity. A slightly outdated technology feature description is lower.

FAILURE CATEGORY 3, CONFIDENT OVER-GENERALISATION What it is: AI produces statements that are broadly plausible but overstate how generally applicable they are, missing important exceptions, jurisdictional variations, sector-specific differences, or situational nuances.

Why it happens: AI generates the most generally accurate response to the most common interpretation of the question. It doesn't automatically surface the ways in which the answer differs for your specific situation.

Where it appears most:

Legal questions with jurisdictional variation ("employment law says X" without specifying which country's) Medical questions with individual variation ("the standard treatment is X") Business advice without sector or context sensitivity Technical guidance where the "standard" approach isn't appropriate for specific circumstances

How to catch it: Ask explicitly: "In what situations might this advice not apply? What are the important exceptions?" For professional matters, verify against advice calibrated to your specific jurisdiction and circumstances.

FAILURE CATEGORY 4, ARITHMETIC ERRORS What it is: AI makes errors in arithmetic, calculation, and numerical reasoning, particularly on multi-step calculations, percentage calculations, and anything requiring exact



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arithmetic.

Why it happens: AI generates text. Numbers are text. When AI produces "the answer is 43%," it has generated the number 43 because it is plausible in context, not because it has calculated and arrived at 43. Text generation and mathematical calculation are different processes. Without code execution, AI maths is unreliable.

Where it appears most:

Percentage calculations in reports or analyses Financial modelling or estimation Multi-step arithmetic Unit conversions Time-based calculations (especially with complex date arithmetic)

How to catch it: Check all arithmetic independently. Use code execution when available for anything that matters. Never use an AI-generated financial figure without verifying the arithmetic.

The severity: High when numbers appear in decisions, reports, or presentations. A wrong percentage in a financial summary is a professional error regardless of where it came from.

FAILURE CATEGORY 5, CONTEXT DRIFT What it is: In long conversations, AI gradually loses track of constraints and context established early in the conversation, reverting to defaults it would apply without those constraints.

Why it happens: AI attends to the full conversation but with diminishing precision as length increases. Constraints stated in message one of a twenty-message conversation may be less firmly applied at message eighteen than they were at message two.

Where it appears most:

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Long editing sessions where style constraints gradually slip Extended research conversations where early framing becomes less influential Multi-stage workflows where the first stage's constraints should carry forward but don't

How to catch it: For long conversations, periodically re-state key constraints. When output seems to have drifted from what you specified, check the early conversation and explicitly re-anchor the

constraint.

FAILURE CATEGORY 6, SYCOPHANCY What it is: AI tends to agree with and validate what the user presents, even when the user is wrong. It will often affirm an incorrect claim rather than challenge it, adjust its previous answer to accommodate the user's pushback even when its original answer was correct, and frame critique in ways that are so mild as to be useless.

Why it happens: AI is trained using human feedback, which tends to reward agreeable responses. This creates a bias toward agreement and validation.

Where it appears most:

When you present a flawed plan and ask AI to review it When you push back on AI's correct assessment and it backs down When you ask AI to evaluate your own writing or thinking

How to catch it: Actively ask for challenge. Use adversarial critique prompts (Chapter 18). Ask "what is wrong with this?" not "what do you think of this?" When AI agrees with you completely, treat that as a flag worth checking, not a validation.

The severity: Moderate. Sycophancy is more a quality issue than a factual one, but it can lead to significant errors when you're looking for honest critique of important work.



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FAILURE CATEGORY 7, INSTRUCTION DRIFT What it is: AI partially follows instructions rather than fully following them. It may honour the spirit of an instruction while missing a specific requirement, or follow all instructions except one.

Why it happens: Complex sets of instructions create competing pressures. AI optimises across them in ways that may not match your priority ordering. An instruction to "be brief" may be honoured by shortening paragraphs while maintaining a section structure you asked to be removed.

Where it appears most:

Multi-constraint prompts with several specific requirements Editing instructions where some changes are made but others are missed Format specifications where the structure is approximately right but details are wrong

How to catch it: For important outputs, review explicitly against your stated constraints. Check each instruction in your prompt: was it followed? If not, restate it as the sole instruction in a follow-up.

FAILURE CATEGORY 8, PLAUSIBLE BUT WRONG REASONING What it is: AI produces reasoning that appears sound, logical structure, coherent argument, sensible-sounding progression, but reaches the wrong conclusion, misapplies a principle, or draws on an incorrect premise.

Why it happens: Plausible reasoning is generated in the same way as plausible text. A well-structured argument is a pattern AI has encountered many times and can reproduce convincingly. The quality of the reasoning structure does not guarantee the quality of the reasoning.

Where it appears most:

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Legal analysis Financial reasoning Causal analysis ("X happened because...") Risk assessment Strategic recommendations

How to catch it: For important analytical output, apply the verification question: "Does the conclusion actually follow from the premises?" Test the reasoning by looking for an alternative explanation or conclusion that the same evidence could support. Have a qualified person review reasoning in high-stakes domains.

PRO TIP WHEN YOU RECEIVE AN OUTPUT THAT MATTERS, ASK AI

"What are the three most important ways this analysis could be wrong? What assumptions am I relying on that might not hold?" This is not a perfect catch-all, but it surfaces failure modes AI can identify, which is often enough to direct your verification effort toward the right areas.



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A PRACTICAL SUMMARY: WHERE TO BE MOST CAREFUL



OUTPUT TYPE PRIMARY FAILURE RISK VERIFICATION PRIORITY

Statistics and Hallucination, arithmetic HIGH, verify all figures numbers error

Citations and Hallucination HIGH, verify all citations references exist

Legal/regulatory Outdated info, over- HIGH, verify from current claims generalisation official sources

Medical information Over-generalisation HIGH, verify from clinical sources

Reasoning and Plausible-wrong MEDIUM, check argument analysis reasoning, sycophancy logic

Current Outdated information MEDIUM, verify recency events/market data

Technology Outdated information MEDIUM, verify from specifications official docs

Writing and structure Instruction drift LOW, read against your brief

Creative framing Generally lower risk LOW, assess on merits

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